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小苏打浓度与煮制时间对意大利面拉面化效果的影响：一项 3 × 3 探索性厨房实验

Effects of Baking Soda Concentration and Cooking Time on the Ramen-Style Transformation of Spaghetti: A 3 × 3 Exploratory Kitchen Trial

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Figure 1. Finished-dish presentation documented by the author. Commercial products were independently purchased.

摘要与关键词 / Abstract and Keywords

摘要： 本探索性厨房实验考察小苏打浓度与煮制时间对商品干制意大利面“拉面化”感官表现的共同影响。实验采用3 × 3处理矩阵，将2、4、6 g/L三种小苏打浓度与10、12、15分钟三种煮制时间组合，共形成九种处理。作者依据弹性、中心熟度、顺滑度、碱香纯度、苦涩感及与拉面风格面条的整体相似度进行结构化描述，并给予10分制综合评分。结果显示，4 g/L、15分钟处理得分最高（9.5/10），表现为颜色金黄、面体顺滑、弹性较好且碱香适中；4 g/L、12分钟得分8.5/10，保留较明显的偏硬中心。2 g/L处理的碱性特征总体较弱，而6 g/L处理随时间延长出现苦味、涩感、表面黏口及肥皂样气味。该结果支持将4 g/L、15分钟作为家庭厨房试制的实用起点，但由于研究为单人评价、未报告独立重复且缺少pH、仪器质构和烹煮损失测量，结论应理解为透明的探索性观察，而非统计学推广结果。

Abstract: This exploratory kitchen trial examined the combined effects of baking-soda concentration and cooking time on the ramen-style transformation of commercial dried spaghetti. A 3 × 3 treatment matrix combined three concentrations, 2, 4, and 6 g/L, with cooking times of 10, 12, and 15 minutes, producing nine treatments. The author documented elasticity, center doneness, surface smoothness, alkaline aroma, bitterness or astringency, and overall similarity to ramen-style noodles, then assigned a descriptive overall score on a 10-point scale. The highest score, 9.5/10, was assigned to 4 g/L for 15 minutes. This treatment was described as golden, smooth, springy, and appropriately alkaline without obvious bitterness. The 4 g/L, 12-minute condition scored 8.5/10 and retained a firmer center. Treatments at 2 g/L produced limited alkaline character, whereas prolonged cooking at 6 g/L produced bitterness, astringency, surface stickiness, and a soap-like odor. The findings support 4 g/L for 15 minutes as a practical starting condition for home use, but the results remain exploratory because the work involved one evaluator, undocumented replication, and no pH, instrumental texture, or cooking-loss measurements.

关键词 / Keywords:
小苏打；意大利面；碱水面；拉面替代；感官评价；厨房实验 / baking soda; spaghetti; alkaline noodles; ramen substitute; sensory evaluation; culinary methods

1. 研究背景 / Background

在部分地区，新鲜碱水面或成品拉面的获取并不稳定，而干制意大利面具有价格可控、规格稳定和储存方便等特点。小苏打进入煮面水后会提高体系碱性，可能影响面条颜色、气味、表面结构和面芯熟化速度。网络烹饪经验常建议“加一点小苏打”来制造拉面感，但浓度和时间通常缺乏系统比较。

Fresh alkaline noodles are not consistently available in every market, whereas dried spaghetti is widely distributed, shelf-stable, and manufactured to a relatively consistent specification. Adding baking soda raises the alkalinity of the cooking medium and may alter noodle color, aroma, surface properties, and center softening. Informal recipes often recommend adding an unspecified amount of baking soda, but concentration and cooking time are rarely compared in a structured matrix.

本研究的目标是：在家庭厨房条件下，以统一水量建立浓度与时间的交叉比较；记录各处理的主要感官差异；识别最有希望的实用组合；并如实说明证据边界。

The objectives were to compare concentration and time under a common water volume, document the sensory differences among treatments, identify a practical candidate condition, and state the limits of the evidence transparently.

2. 材料与方法 / Materials and Methods

材料。 实验使用商品干制硬质小麦意大利面、小苏打和饮用水。包装标示常规煮制时间约10-11分钟。原始图片记录了4 g小苏打称量，但没有记录每组干面质量。

Materials. Commercial dried durum-wheat spaghetti, baking soda, and water were used. The package indicated a conventional cooking time of approximately 10-11 minutes. A 4 g baking-soda measurement was photographed, but dry pasta mass per treatment was not recorded.



Figure 2. Commercial spaghetti and measurement of 4 g baking soda.

实验设计。 每组使用1 L水。小苏打浓度设置为2、4、6 g/L，煮制时间设置为10、12、15分钟，共九组。实验在家庭炉具上完成，原始记录未提供独立重复次数或随机化顺序。

Design. Each treatment used 1 L water. Concentrations were 2, 4, and 6 g/L, and cooking times were 10, 12, and 15 minutes. The trial was conducted on a household stove. Independent replication and randomized treatment order were not documented.

评价方法。 作者根据面条弹性、表面顺滑度、中心硬度、碱香、苦涩或肥皂样异味，以及与拉面风格面条的整体相似度进行结构化描述，并给予综合评分。评分属于单人主观判断，并非训练型感官小组或盲评。

Evaluation. The author recorded springiness, surface smoothness, center firmness, alkaline aroma, bitterness or soap-like odor, and overall ramen-style similarity. The score represents one evaluator’ s structured judgment rather than a blinded or trained sensory panel.

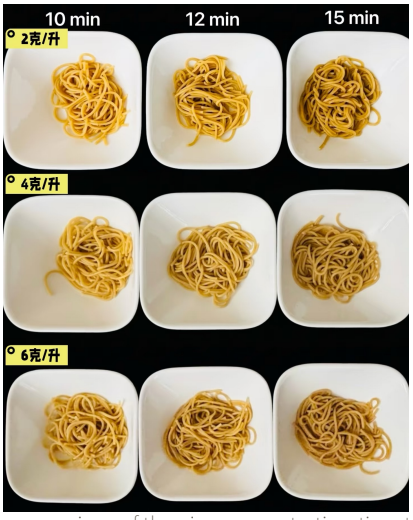


Figure 3. Visual comparison of the nine concentration-time treatments.

3. 结果 / Results

Baking soda	10 min	12 min	15 min
2 g/L	5.0	6.0	6.5
4 g/L	7.0	8.5	9.5
6 g/L	6.0	5.0	3.5

2 g/L组随时间延长仅呈温和改善：10分钟仍接近普通意大利面，12分钟碱香不足，15分钟虽熟透但偏软。4 g/L组呈持续上升趋势：10分钟外弹内硬，12分钟更接近偏硬拉面，15分钟获得最佳综合表现。6 g/L组则随时间延长持续下降：10分钟已出现偏强碱味，12分钟苦涩和黏口，15分钟出现明显肥皂样气味和苦味。

At 2 g/L, scores improved only modestly with time: the 10-minute sample remained spaghetti-like, the 12-minute sample had weak alkaline character, and the 15-minute sample was fully cooked but soft. At 4 g/L, scores improved across time: the 10-minute treatment had an elastic exterior but firm center, 12 minutes approached a firmer ramen style, and 15 minutes produced the best overall result. At 6 g/L, scores declined with time as bitterness, astringency, stickiness, and soap-like odor intensified.



Figure 4. Original score matrix and treatment descriptions.

4. 实用方法 / Practical Procedure

推荐起始方法为：向1 L水中加入4 g小苏打并煮沸；加入意大利面后继续煮15分钟；使用容量较大的锅以应对泡沫；捞出后用于汤面或拌面。偏好更硬面芯时可将时间缩短至约12分钟。该方法应被视为起始条件，实际时间仍会受到面条直径、炉具功率、海拔和品牌配方影响。

A practical starting method is to add 4 g baking soda to 1 L water, bring the solution to a boil, add spaghetti, and cook for 15 minutes. A larger pot helps manage foaming. The noodles may then be used in soup or mixed-noodle preparations. A 12-minute endpoint may suit a firmer preference. Required time can vary with strand diameter, stove output, altitude, and product formulation.

5. 讨论 / Discussion

实验结果最重要的模式并非“碱越多越好”，而是中等浓度与充分熟化之间的平衡。2 g/L可能不足以产生明显的碱水面特征；6 g/L则容易在延长煮制时积累令人不悦的苦涩和肥皂样风味。4 g/L提供了较大的可操作窗口，在12至15分钟之间可以通过时间调节面芯硬度。

The central pattern was not that more alkali produced a better noodle. Instead, a middle concentration had to be balanced with sufficient center softening. At 2 g/L, alkaline character was limited. At 6 g/L, prolonged cooking amplified undesirable flavor and surface properties. The 4 g/L condition created a usable window in which cooking time could be adjusted between approximately 12 and 15 minutes to change center firmness.

照片中的颜色随浓度和时间总体加深，但该变化不能被当作仪器色差结果。相机曝光、白平衡、拍摄时间和样品表面水分均可能影响视觉判断。

Color appeared to darken with concentration and time, but the photographs are not instrumental color measurements. Exposure, white balance, time after cooking, and surface moisture could all influence appearance.

6. 局限性 / Limitations

本研究未报告每组干面质量、独立重复次数、随机化与盲评流程，也没有独立评审人数。10分制综合评分未经统计验证。未测量pH、含水率、烹煮损失、断裂力、拉伸强度或仪器色差。普通水煮意大利面虽被提及为概念性对照，但未提供独立评分。因此，本研究应被定位为可供复现实验设计参考的探索性厨房报告。

Dry pasta mass, independent replication, randomized order, blinding, and the number of assessors were not documented. The 10-point score was not statistically validated. pH, moisture, cooking loss, breaking force, tensile properties, and instrumental color were not measured. A conventional-water control was mentioned but not scored separately. The study should therefore be interpreted as an exploratory culinary report that can inform a more rigorous replicated experiment.

7. 结论 / Conclusion

在本次九组探索性比较中，4 g/L小苏打、15分钟处理获得最高评分，4 g/L、12分钟提供更硬的替代选择。高浓度和长时间组合表现最差。后续研究应增加重复实验、独立感官评价、pH和仪器质构测量，并记录干面质量与烹煮损失。

Among the nine treatments, 4 g/L for 15 minutes received the highest score, while 4 g/L for 12 minutes provided a firmer alternative. High concentration combined with prolonged cooking performed worst. Future work should add replication, independent sensory assessment, pH and instrumental texture measurements, dry-mass standardization, and cooking-loss analysis.

出版信息 / Publication Information

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